

Rice Securities Investment Group
Liquidia Corp (NASDAQ: LQDA)
Biotechnology

Background & Summary

Liquidia Corporation is a biopharmaceutical company specializing in creating cutting-edge therapies for pulmonary arterial hypertension (PAH) and pulmonary hypertension associated with interstitial lung disease (PH-ILD). PAH and PH-ILD are chronic diseases where high blood pressure in the lungs stresses the heart's ability to pump blood, caused respectively by narrow arteries and lung disease. Both conditions are serious, progressive, and currently incurable, requiring long-term pharmacologic management. The incumbent industry leader, United Therapeutics (NASDAQ: UTHR), launched Tyvaso (nebulizer) in 2009 and Tyvaso DPI (dry-powder) in 2022. Today, the U.S. addressable population includes approximately 45,000 PAH and 60,000 PH-ILD patients, roughly \$4.1 billion in annual net revenue (2024). LQDA's lead product, Yutrepia (dry-powder), became the second therapy approved for both conditions, utilizing the proprietary PRINT technology. Since its FDA approval in May 2025, YUTREPIA has achieved a trajectory rarely seen in specialty pharma launches, generating \$148.3M in net sales in just seven months, with a Q4 2025 annualized run rate of ~\$360M.

Liquidia is a clear example of a market inefficiency where legacy fears overshadow current fundamental reality. The stock is trading at a valuation that seems to ignore the explosive commercial launch of YUTREPIA (12x P/E for a company growing QoQ revenue by nearly 100%). The market discounts Liquidia on three perceived overhangs: the announcement of UTHR's TreSMI, the ongoing patent litigation with UTHR, and the emerging threat of Insmed's TPiP. We believe the market's concerns are overstated. Liquidia is steadily taking share from UTHR, the remaining litigation risk should be resolved in Liquidia's favor, and the company's accelerated timeline and continued commercial execution will drive a meaningful re-rating for the stock.

Investment Thesis

1. YUTREPIA's superior dosing flexibility and tolerability are driving best-in-class commercial, allowing it to gradually capture share from United Therapeutics despite the market's belief that Tyvaso will remain a dominant therapy

The market continues to view UTHR as the entrenched incumbent in inhaled prostacyclins, assuming Tyvaso will remain the dominant therapy despite the launch of YUTREPIA due to 1) high switching costs and 2) payer access. UTHR has been the standard of care in inhaled prostacyclins for over a decade through Tyvaso, which already has thousands of patients on therapy. Moreover, UTHR has had years to negotiate payer contracts and reimbursement pathways for Tyvaso, leading investors to grow skeptical on YUTREPIA's market penetration. However, early commercial data suggests the opposite dynamic unfolding: YUTREPIA is rapidly gaining traction and appears to be taking share from legacy therapies such as Tyvaso.

Liquidia's competitive moat is tied to its proprietary PRINT (Particle Replication in Non-wetting Templates technology). Historically, dry powder inhalers (DPIs) have struggled with variable particle sizes, leading to inconsistent lung deposition and cough. PRINT technology allows Liquidia to engineer drug particles with uniform size, shape, and chemical composition. This uniformity is not cosmetic, rather functional. By creating particles that are aerodynamically optimized for deep lung delivery, YUTREPIA is able to achieve higher doses (strongly correlated with improved outcomes) with a convenient, low-effort inhaler.

Despite this, recent market concern about Liquidia has centered around UTHR's announcement of TreSMI, a soft-mist inhaler designed to reduce coughing associated with inhaled Treprostinil therapies. UTHR claimed that "human studies" showed cough reduction by up to 90% and that they intend to file this drug for approval this year and launch

Investment Summary

Recommendation	LONG
Price Target	\$53.36
Current Price	\$31.02
Implied Upside	72.03%
Timeline	18 Months
52-Week Range	\$11.26 - \$46.67

Market Data

Market Cap	\$2,696.6M
Enterprise Value	\$2,740.3M
EV/Sales	39x
P/E	12x

Income Statement LTM

Revenue	\$69.2M
Yutrepia (PAH)	\$39.4M
Yutrepia (PH-ILD)	\$26.4M
R&D	\$38.8M
SG&A	\$130.1M
Net Income	-\$121.8M

Balance Sheet

Cash & Equivalents	\$176.5M
Net Debt	-\$169.4M

Cash Flow Statement LTM

Operating Cash Flow	-\$100.7M
Free Cash Flow	\$107.8M

Price Chart - 1Y



commercially next year. However, no supporting clinical data, head-to-head evidence versus existing therapies, or trial timelines were given. They also announced the development of a super prostacyclin which would last longer and allow for higher dosages. Essentially, this is an attempt from UTHR to release 2 products that address concerns that YUTREPIA has already addressed more efficiently. The market misses that mist inhalers are not new and are less efficient than dry powder inhalers due to improper titration. Secondly, even if TreSMI does launch, it would certainly take more than the year that UTHR claims. For some context, Tyvaso DPI took 3.5 years to commercialize, and considering the fact that TreSMI has no active clinical studies yet, it would be unprecedented if they are able launch in a year. This means Liquidia has at least a 2-year head start to establish YUTREPIA as the preferred inhaled prostacyclin therapy and continue building a durable prescriber base

Furthermore, there is significant commercial uptake which validates this moat. 25% of YUTREPIA prescriptions are switches from other prostacyclin therapies. The conversion rate of 85% among prescribers suggests that once physicians encounter the PRINT technology, they prefer it over legacy options (e.g., Tyvaso DPI). In fact, Tyvaso DPI has serious tolerability issues: 70% of PH-ILD patients discontinued it within 42 days, while YUTREPIA discontinuation was much lower at 20% and spanned a significantly larger time frame (6 months). In Q4 2025, Tyvaso DPI sales declined \$13.7M while YUTREPIA grew by over \$40M.

2. The street's concerns of UTHR's infringement lawsuit regarding the '327 patent is overblown, and Liquidia is well-positioned to continue commercializing YUTREPIA even in the unlikely chance that they are unsuccessful in the lawsuit

Liquidia and UTHR have a deep history of legal warfare, with UTHR filing multiple lawsuits prior to and following the FDA approval of YUTREPIA:

- 2018: UTHR files a trade secret lawsuit against Liquidia → Ruled in favor of Liquidia
- 2019: UTHR sues Liquidia claiming YUTREPIA infringes the '793 patent (use of inhaled treprostinil treating pulmonary hypertension) → PTAB invalidates and decision is affirmed on appeal, removing biggest barrier to YUTREPIA
- 2023: UTHR asserts '327 patent (methods of treating PH-ILD using inhaled treprostinil) → Final ruling still pending
- 2024: UTHR requests for preliminary injunction (PI), which would block YUTREPIA from launching → Denied
- 2025: UTHR also pushes the '782 patent (continuation of invalidated '793 patent with minor changes) → Final ruling still pending

The '782 patent is essentially a regurgitation of the already invalidated '793 patent with one minor claim element (dosing timing). Data suggests that patentees who fail to secure a PI win on the merits only ~20-25% of the time. Judge Andrews specifically said "Plaintiffs have not shown a likelihood of success on the merits," which is the very first test/requirement for a PI to be granted and suggests that the court did not believe UTHR had a strong case that Liquidia infringes the patent. Adjusting for the previous invalidation of the family patent, the probability that Liquidia prevails is ~80-85%. The market, however, is pricing it like a 50/50 coin-flip, which is a clear mispricing of the legal probability.

Sell-side equity analysts claim that they "lean in the direction of Judge Andrews likely finding that at least one asserted Claim of '327 Patent is Valid and Infringed". However, there is compelling evidence to support its denial from previous trials that Judge Andrews ruled on. The Bayer v. Mylan (2025) Federal Circuit decision held that pharma patients cannot claim clinical trial outcomes along without specifying the method that produces those results. This is relevant to the UTHR '327 patent which includes many dependent claims tied to measurable outcomes (e.g. improving 6MWD, NT-proBNP, FVC). During the case, Judge Andrews asked both parties to brief the Bayer ruling, which is very rare in cases like these and signals that he may view those outcome-based claims as vulnerable under this precedent. If those dependent claims were weakened or invalidated, UTHR would be left relying on broader independent claims that are harder to enforce and easier for Liquidia to defend. Moreover, the fact that YUTREPIA is already on the market changes the commercial urgency of this case as any ruling on invalidity or non-infringement has an immediate impact on patients and prescribers. For some context, there is no known case where a drug approved by the FDA has been removed from market solely due to patent litigation in the U.S. Bears argue that despite a win, LQDA would face appeals, commercial barriers and delaying tactics from United Therapeutics, but a legal victory would give LQDA enough runway to build meaningful market share and continue disrupting Tyvaso sales.

There are three main scenarios if the trial outcome is negative (although Judge Andrews' written opinion will only state who won, and the consequences of the decision will be addressed in a subsequent step into the future):

- 1) Royalty payments (least severe): Liquidia keeps selling YUTREPIA but pays UTHR a royalty on sales (most common remedy)
- 2) Remove PH-ILD indication from label (moderate outcome): Liquidia removes the PH-ILD indication from YUTREPIA's FDA Label
- 3) YUTREPIA removed from the market (worst case): Extremely unlikely considering YUTREPIA is approved and prescribed

3. While YUTREPIA drives near-term cash flow and market penetration, the street is discounting L606's structural dosing advantage and PH-ILD's potential to surpass PAH as the company's primary long-term growth driver

The development of L606 is a natural extension of Liquidia's technological moat and represents a structural expansion of the prostacyclin franchise. The primary limitation of current inhaled treprostinil therapies is QID dosing (4x a day), which YUTREPIA and Tyvaso currently require. On the other hand, L606, a liposomal formulation of treprostinil currently in the global Phase 3 Re-Spire trial, is designed for BID dosing (2x a day). This doesn't just make the use of the drug more convenient, but improves the quality and efficiency of the treatment through two main factors: (1) reduced

peak-to-trough pharmacokinetic variability, resulting in more consistent pulmonary vasodilation and hemodynamic stability, and (2) improved tolerability driven by lower peak systemic exposure (C_{max}), which may enhance adherence and long-term persistence. Phase 1 data demonstrated comparable overall exposure (AUC) with approximately 7x lower C_{max} versus Tyvaso, alongside materially lower cough incidence, suggesting improved tolerability while preserving therapeutic exposure (Figure 1). Moreover, 48-week data showed minimal peak-to-trough differences in 6-minute walk distance (6MWD), reinforcing the thesis that sustained exposure may translate into more stable hemodynamic control. The Re-Spire trial evaluates both peak and trough 6MWD in 344 PH-ILD patients, directly testing this hypothesis. Most importantly, treprostinil's efficacy in PH-ILD has already been validated through the INCREASE trial, and given L606's reformulation pathway under 505(b)(2), clinical risk is materially lower than that of a novel drug. Furthermore, the development of the L606 drug is complemented by Liquidia's PRINT technology, as it would be impossible to create with standard techniques. As shown in our model, L606 PH-ILD revenue is projected to ramp from ~60M at launch to a peak of ~775M. Overall, the street mainly disregards the differentiated durability of L606, effectively treating Liquidia as a single-asset YUTREPIA litigation story rather than a continuous evolution of their prostacyclin platform.

On the other hand, PH-ILD shows a structurally unpenetrated growth market beyond PAH gains. While most PAH patients are already treated and prescribed, the majority of PH-ILD patients are untreated. The U.S. PH-ILD market represents an addressable population of over 60,000 patients, yet with only ~750 active prescribers out of a target universe of 6,500 (around 11.5% penetration), this segment offers substantial runway for growth. Clinical validation from INCREASE established inhaled treprostinil as an effective therapy in PH-ILD, and subsequent ASCENT data for YUTREPIA demonstrated continued dose titration with sustained improvements in 6MWD and manageable tolerability, reinforcing physician confidence in higher dosing strategies. As diagnosis rates improve and community pulmonologists grow more comfortable initiating inhaled prostacyclins, PH-ILD has the potential to eclipse PAH as the primary revenue driver of the franchise, with our model projecting PH-ILD revenue expanding from ~\$229M in 2026 to a peak of ~\$1.17B by 2032, nearly 2x the size of the projected PAH segment (~\$664M) at peak. When combined with Liquidia's recent clinical studies that target inadequate responders and adjacent indications such as IPF, PPF, PH-COPD, and SSc-RP, Liquidia's long-term value proposition shifts from a one-time share price litigation rerate versus United Therapeutics to a durable cardiopulmonary platform with multi-year expansion strategies.

Catalysts

1. **'782 and '327 patent case decision:** The U.S. District Court for the Middle District of North Carolina and District of Delaware is scheduled to release a ruling for the '782 and '327 patent by mid-2026 based on historical timelines. Following previous court rulings in favor of LQDA, such as the May 30, 2025 ruling to deny UTHR's Temporary Restraining Order and Preliminary Injunction due to substantial uncertainty about the validity of UTHR's original '782 patent and a lack of proof of irreparable harm to UTHR, litigatory odds are in LQDA's favor. According to historical records on U.S. patent litigation, patentees who fail to secure a preliminary injunction only win on merits 20-25% of the time. Combined with the previous ruling to deny UTHR's assertion of LQDA's infringement on UTHR's '793 patent in the same patent family in July 2022, we believe there is a compelling 80-85% probability LQDA to win the '782 and '327 patent case. A ruling in favor of LQDA would clear the final legal overhang for LQDA's continued launch of YUTREPIA.
2. **Initiation of phase 3 RESPIRE trials for L606:** LQDA is planning to initiate their phase 3 trials of L606 in mid-2026, which will provide necessary data needed for FDA regulatory approval and give LQDA an edge against its competitors. Various factors already create a compelling case for L606's approval and successful launch. UTHR's successful results from the INCREASE trials in February 2020 have validated the efficacy of Treprostinil to treat PH-ILD patients. Additionally, estimates for L606's probability of success to proceed from phase 3 of testing to market authorization are favorable, at around a 60%. If L606 gets FDA approved, it would be a massive win for Liquidia due to the drug's key competitive advantages such as 2x (BID) daily dosing as opposed to YUTREPIA and Tyvaso DPI's 4x (QID) daily dosing.

Risks

1. **Injunction on '782 and '327 patent:** LQDA faces uncertainty regarding the decision on its alleged infringement on UTHR's '782 and '327 patents. While unlikely, the court could issue a permanent injunction, which would halt YUTREPIA sales and hurt LQDA's share price. However, the likelihood of the injunction being successful is estimated to be <20% due to historical rulings on LQDA and general patent litigation. Previously, the court has already found that the extent of the erosion of UTHR's prices is speculative, especially in a market where an increasing number of patients are being diagnosed with relevant illnesses. Additionally, the court has found that that YUTREPIA is not direct substitute to Tyvaso, thus rendering it invalid as a direct threat to UTHR. Even in the court rules in favor of UTHR, no drug has ever been removed from the market solely due to patent litigation, and we believe the most likely scenario would be LQDA paying UTHR a royalty on YUTREPIA sales.
2. **Competition from other biopharmaceutical companies:** Despite LQDA's competitive moat due to its PRINT technology, high dosage capacity, and early start on drug development, it faces competition from significant competitors (most notably Insmed's TPIP and Merck's Winrevair). Insmed's development of TPIP addresses challenges, such as dosing rates, that Tyvaso and YUTREPIA face, as it only requires dosing 1x/day and demonstrated lower pulmonary vascular resistance (PVR) by 35% in the June 2025 phase 2b study, and offers significant improvements for patients' quality of life. However, even if TPIP gets FDA approved, YUTREPIA will have had nearly 4-5 years of commercial incumbency as TPIP's Phase 3 results are expected by 2028/2029. Merck's Winrevair (which is already FDA approved) is differentiated due to its disease modifying quality, as the drug also targets the underlying causes of PAH such as vascular remodeling and cellular proliferation rather than focusing on addressing blood vessel constriction like a traditional vasodilator. However, Winrevair is designed as an add on to PAH medications, making it a complement as opposed to a direct competitor with YUTREPIA as prescriber often bundle the products. Thus, LQDA's competitive edge remains, as its biggest competitors continue to be dampened by YUTREPIA's accelerated timeline and product advantages.

Model

	2024	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
YUTREPIA (Base Case)		2025	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035	2036	2037	2038	2039	2040
Yutrepia - U.S. Sales		148.3	518.1	714.5	948.1	1193.7	1447.2	1658.2	1834.7	1820.0	1780.0	1700.0	1600.0	1480.0	1350.0	1220.0	1100.0
Yutrepia - U.S. PH-ILD		63.3	229.2	353.7	521.5	706.8	898.7	1047.7	1170.3	1180.0	1160.0	1120.0	1050.0	980.0	900.0	820.0	740.0
Yutrepia - U.S. PAH		85.0	288.9	360.8	426.6	486.9	548.5	610.5	664.4	640.0	620.0	580.0	550.0	500.0	450.0	400.0	360.0
Total Revenue	14	159.2	518.1	714.5	948.1	1193.7	1447.2	1658.2	1834.7	1820.0	1780.0	1700.0	1600.0	1480.0	1350.0	1220.0	1100.0
COGS	5.9	16.3	51.8	71.4	94.8	119.4	144.7	165.8	183.5	182.0	178.0	170.0	160.0	148.0	135.0	122.0	110.0
Gross Profit	8.1	142.9	466.3	643.1	853.3	1074.3	1302.5	1492.4	1651.2	1638.0	1602.0	1530.0	1440.0	1332.0	1215.0	1098.0	990.0
R&D	47.8	33.3	108.3	81.5	89.1	93.1	98.4	97.8	99.1	98.3	96.1	91.8	86.4	79.9	72.9	65.9	59.4
SG&A	81.6	150.9	491.2	273.7	305.3	319.9	338.6	353.2	376.1	373.1	364.9	348.5	328.0	303.4	276.8	250.1	225.5
Total Operating Expenses	129.4	184.3	599.4	355.1	394.4	413.0	437.1	451.0	475.2	471.4	461.0	440.3	414.4	383.3	349.7	316.0	284.9
Operating Income	-121.3	-41.4	-133.2	287.9	458.9	661.3	865.4	1041.3	1176.0	1166.6	1141.0	1089.7	1025.6	948.7	865.4	782.0	705.1
Effective Tax Rate %	20.00%	0.2	0.2	0.2	0.2	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3
NOPAT		-33.1	-106.5	230.4	367.1	496.0	649.1	781.0	882.0	875.0	855.7	817.3	769.2	711.5	649.0	586.5	528.8
Discounted Cash Flows		-33.1	-95.1	183.6	261.3	315.2	368.3	395.7	399.0	353.4	308.6	263.1	221.1	182.6	148.7	120.0	96.6
WACC		12%															
Growth rate		2%															
NPV of Cash Flows		3489.1															
NPV of Terminal Value		985.5															
Total NPV		4474.6															
rNPV		3803.4															
Litigation Risk Adjustment		85.0%															
L606 (PH-ILD)																	
Revenue						60.0	150.0	300.0	480.0	650.0	775.0	760.0	720.0	650.0	550.0	420.0	280.0
Operating Income						35.6	93.8	191.7	308.6	418.0	498.3	488.7	463.0	418.0	353.7	270.1	180.0
NOPAT						26.7	70.3	143.8	231.5	313.5	373.7	366.5	347.2	313.5	265.2	202.5	135.0
Discounted Cash Flows						17.0	39.9	72.8	104.7	126.6	134.8	118.0	99.8	80.5	60.8	41.4	24.7
NPV of Cash Flows						921.0											
NPV of Terminal Value						251.6											
Total NPV						1172.6											
rNPV						728.2											
PoS (Probability of Success)						0.6											
Base Phase 3 Rate						60%											
MoA Validation (from INCREASE trial)						1.15											
Novel Formulation Risk						0.9											

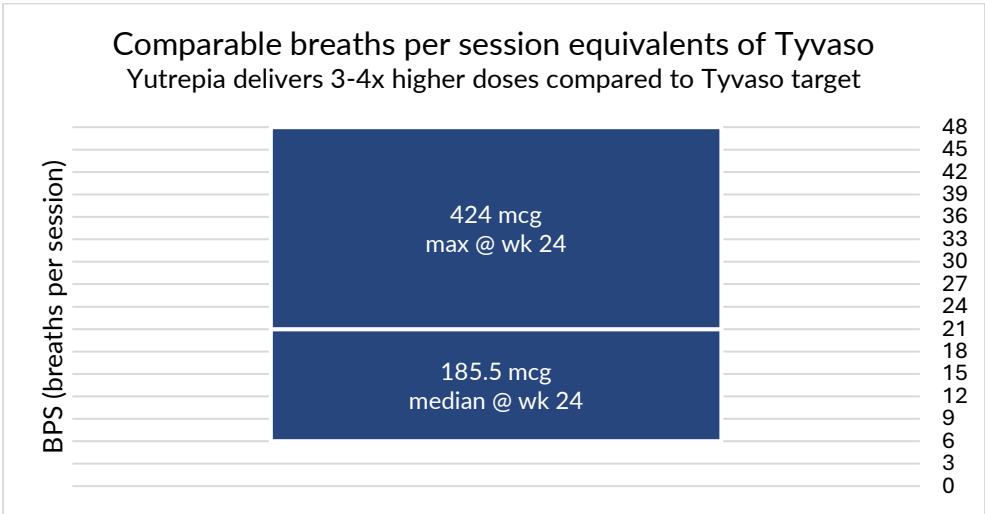
Generic Treprostinil																	
Revenue		10.9	11.3	11.4	11.5	11.7	11.8	11.9	12.0	12.0	12.0	12.0	12.0	12.0	12.0	12.0	12.0
Operating Income		2.7	2.8	2.9	2.9	2.9	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0
NOPAT		2.2	2.3	2.3	2.3	2.3	2.4	2.4	2.4	2.4	2.4	2.4	2.4	2.4	2.4	2.4	2.4
Discounted Cash Flows		2.2	2.0	1.8	1.6	1.5	1.3	1.2	1.1	1.0	0.9	0.8	0.7	0.6	0.6	0.5	0.4
NPV of Cash Flows		18.2															
Liquidia SOTP Valuation	rNPV (\$M)	Per Share (\$)															
YUTREPIA	3803.39	43.82															
L606 (PH-ILD)	728.20	8.39															
Generic Treprostinil	18.16	0.21															
Corporate Overhead	-90.00	-1.04															
Total EV	4459.75	51.38															
Net Cash	43.00	0.50															
Total Equity Value	4502.75	\$ 51.88															
Shares Outstanding (M)	86.8																
Expense Ratios and Margins																	
Gross Margin - U.S. Yutrepia Sales		90.00%	90.00%	90.00%	90.00%	90.00%	90.00%	90.00%	90.00%	90.00%	90.00%	90.00%	90.00%	90.00%	90.00%	90.00%	90.00%
Gross Margin -Other Sales		60.00%	60.00%	60.00%	60.00%	60.00%	60.00%	60.00%	60.00%	60.00%	60.00%	60.00%	60.00%	60.00%	60.00%	60.00%	60.00%
Gross Margin - Total Revenues		89.80%	89.40%	89.50%	89.60%	89.70%	89.80%	89.80%	89.80%	89.80%	89.80%	89.80%	89.80%	89.80%	89.80%	89.80%	89.80%
Selling, General & Administrative		582.80%	94.80%	38.30%	32.20%	26.80%	23.40%	21.30%	20.50%	20.50%	20.50%	20.50%	20.50%	20.50%	20.50%	20.50%	20.50%
Research & Development		341.80%	20.30%	11.40%	9.40%	7.80%	6.80%	5.90%	5.40%	5.40%	5.40%	5.40%	5.40%	5.40%	5.40%	5.40%	5.40%
Operating Margin		25.00%	39.60%	48.00%	55.10%	59.40%	62.50%	63.90%	64.30%	64.30%	64.30%	64.30%	64.30%	64.30%	64.30%	64.30%	64.30%

	Probability	Price
Bull Case	25%	\$ 71.44
Base Case	55%	\$ 51.88
Bear Case	20%	\$ 34.86
Probability-Weighted Price Target		\$ 53.36
Current Price		\$ 31.02
Implied Upside		72.03%

Price Target Sensitivity Analysis (Litigation Risk Adjustment / Peak Sales in \$M)			
	1376	1835	2293
75% (High Risk)	34.86	46.72	58.53
85% (Moderate Risk)	38.72	51.88	64.98
95% (Low Risk)	42.58	57.03	71.44

Appendix

Figures for Thesis I



Figures for Thesis III

